

# Outage Investigation Report —

## OIR-2025-008

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| Field             | Value                                                                           |
|-------------------|---------------------------------------------------------------------------------|
| Document ID       | OIR-2025-008                                                                    |
| Report Date       | 2025-11-24                                                                      |
| Author            | Davis Ngo, Reliability Engineer I                                               |
| Reviewing Manager | Priya Ramachandran, Plant Manager, NW-PowerPool                                 |
| Region            | NW-PowerPool                                                                    |
| Site              | NW-PowerPool-WIND-024                                                           |
| Asset             | SUB-10629 (Hitachi Energy, substation_equipment)                                |
| Outage ID         | OUT-2025-1600                                                                   |
| Classification    | Forced outage — soc_drift                                                       |
| Distribution      | NW-PowerPool Operations, Reliability Engineering, Vendor Management, Compliance |

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## 1. Outage Summary

SUB-10629 declared a forced outage on 2025-11-12 at 03:42 PST on protection action triggered by SoC drift on collector battery buffer Bank B-3. The asset returned to service on 2025-11-16 at 14:42 PST, total duration **107 hours**. Lost generation: **2,609.0 MWh**.

This is the second SoC-drift outage on SUB-10629 in FY2025; the April event is documented in OIR-2025-001. The event sits in the middle of the Fall Overheating Sequence (October–November); RCA-2025-002 traces the broader pattern.

## 2. Timeline of Events

| Timestamp (PST)            | Event                                                                                                                                                                                                                                          |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2025-11-11<br>22:00        | SoC drift on Bank B-3 begins climbing from baseline 0.6% to 1.8% across 4 hours.                                                                                                                                                               |
| 2025-11-12<br>02:55        | Drift crosses 2.0% advisory under CSM-7 §4.2. Operator (T. Okafor) is responding to a separate alarm on WIN-10011 (efficiency_decline trend) — see OIR-2025-009 for the WIN-10011 event 20 days later.                                         |
| 2025-11-12<br>03:42        | Protection trip on under-voltage co-incident with drift at 3.2%. Forced outage declared.                                                                                                                                                       |
| 2025-11-12<br>05:00        | CPS dispatched; ETA 4 h 10 min. EPA-2024-11 threshold marginally exceeded; on-call manager elected to wait for CPS, citing OIR-2025-001 as precedent for similar work scope and to control cost. See EMR-2025-006 dissent memo.                |
| 2025-11-12<br>09:20        | CPS onsite. BMS Bank B-3 firmware confirmed 2.7.4 — the upgrade was applied in April under OIR-2025-001 closure. New failure mode: thermal management board (TMB) Rev. A fault, distinct from the firmware defect that caused the April event. |
| 2025-11-13 —<br>2025-11-15 | TMB replacement (Hitachi part number HE-TMB-2200-B, supersedes Rev. A), full Bank B-3 calibration, relay re-coordination.                                                                                                                      |
| 2025-11-16<br>14:42        | Asset returned to service.                                                                                                                                                                                                                     |

## 3. Telemetry Observations

- SoC drift at trip: 3.2% (advisory 2.0%, action 2.5% per MS-2025-03 §6.2).
- Bus voltage on 34.5 kV side dropped from 34.1 kV to 32.0 kV in final 75 seconds.
- Cell temperature spread within Bank B-3: 14.4 °C (normal < 5).
- TMB Rev. A fault: voltage reference drift on the temperature sensor input — Hitachi Service Bulletin SB-HE-2025-04 (issued July 2025) recommended proactive replacement with Rev. B for assets > 18 months in service. SUB-10629's TMB was 28 months in service.
- Hitachi SB-HE-2025-04 was logged in the Cascadia OEM bulletin intake but no work order was generated. See CAR-2025-009 (OEM bulletin intake-to-WO gap).

## 4. Restoration Activities

| Vendor                              | Scope                                                                                   | Invoice (USD)    |
|-------------------------------------|-----------------------------------------------------------------------------------------|------------------|
| Cascadia Power Services (CPS)       | Onsite diagnostics, TMB Rev. B replacement, Bank B-3 calibration, relay re-coordination | \$108,400        |
| Hitachi Energy (OEM remote support) | Firmware reconciliation, factory acceptance test sign-off                               | \$34,200         |
| Total                               |                                                                                         | <b>\$142,600</b> |

For comparison, the April event on the same asset (OIR-2025-001) used Apex under EPA and invoiced \$450,500. Roughly a 3.2× ratio for similar scope. See RCA-2025-001 cost arbitrage discussion and EMR-2025-006.

## 5. Preliminary Root Cause

Direct cause: TMB Rev. A voltage-reference drift on the temperature sensor input, causing the BMS to mis-balance Bank B-3 and propagating SoC drift to the protection scheme.

Contributing causes:

1. Hitachi Service Bulletin SB-HE-2025-04 (July 2025) was logged in the OEM bulletin intake but no proactive WO was generated. Eight other NW-PowerPool TMBs are in the same > 18-month-in-service population. See CAR-2025-009.
2. Concurrent alarm pressure on the operator. T. Okafor was triaging both this asset and WIN-10011 trending data. CAR-2025-001 (FMP-2 fatigue) and CAR-2025-006 (CSM-7 procedural shortcut) are relevant.

## 6. Recommendations

1. Mobilize fleet-wide TMB Rev. A replacement at NW-PowerPool within 60 days. Track under CAR-2025-009.
2. Formalize a 30-day SLA on OEM service bulletin intake-to-WO. Track under CAR-2025-009.
3. Document this event's cost benchmark (\$142,600 with CPS) against OIR-2025-001 (\$450,500 with Apex) for vendor MSA renegotiation. Refer to EMR-2025-006.
4. Cross-reference OIR-2025-001 (April SUB-10629 event), OIR-2025-009 (December WIN-10011 event 20 days later) and RCA-2025-002 (Fall Overheating Sequence).

## Signatures

| Role           | Name                              | Date       |
|----------------|-----------------------------------|------------|
| Author         | Davis Ngo, Reliability Engineer I | 2025-11-24 |
| Plant Manager  | Priya Ramachandran                | 2025-11-25 |
| Vendor Manager | L. Halverson                      | 2025-11-26 |

**Distribution:** NW-PowerPool Operations · Reliability Engineering · Vendor Management  
· Compliance · CFO Office

**References:** OIR-2025-001 · OIR-2025-009 · CAR-2025-001 · CAR-2025-006 ·  
CAR-2025-009 · EMR-2025-006 · RCA-2025-001 · RCA-2025-002 · Hitachi Service  
Bulletin SB-HE-2025-04 · Standard MS-2025-03 · CSM-7